

VISHWA G S

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Executive Summary

Mechatronics Engineering undergraduate who has taken a ROS2-based autonomous mobile robot from Gazebo simulation to real hardware, working across navigation, perception, and sensor integration. Skilled in Python, C++, Linux, and Nav2/SLAM, with hands-on experience validating robotic systems in the field. Fast learner who thrives in cross-functional robotics teams.

Technical Skills

- **Robotics & Middleware:** ROS2 (Humble), Nav2, SLAM Toolbox, TF2, URDF, Robot Localization
- **Programming:** Python, C++, C
- **Simulation & Visualization:** Gazebo, RViz
- **Robotics & Control:** Differential-Drive Kinematics, Autonomous Navigation, Motion Control, Sensor Fusion, Wheel Odometry
- **Computer Vision:** OpenCV, YOLOv8, Object Detection, Coordinate Transformation
- **Hardware:** Raspberry Pi, Arduino Mega, ESP32, LiDAR, IMU, Wheel Encoders
- **Tools & Platforms:** Ubuntu Linux, Git/GitHub, PyQt6

Projects

Autonomous Mobile Robot (AMR) — Project Developer

- **Full-stack AMR development:** built a differential-drive AMR on ROS2, taking the system from Gazebo simulation to physical hardware, including URDF design, kinematics, and wheel-encoder-based odometry.
- **Navigation & mapping:** integrated RPLiDAR A2 with SLAM Toolbox for real-time mapping/localization, and implemented Nav2 for autonomous path planning and obstacle avoidance.
- **Sensor fusion:** combined LiDAR, IMU, and wheel-encoder data for robust robot state estimation.
- **Monitoring dashboard:** developed a cross-platform (Windows & Linux) PyQt-based GUI for live robot monitoring, map visualization, manual teleoperation, and autonomous-navigation control.
- **Validation:** tested and debugged mapping, localization, and navigation performance on real-world hardware.

Vision-Guided Robotic Manipulation

- Built a camera-based perception pipeline for object detection and 3D (X-Y-Z) coordinate estimation, with camera-to-robot coordinate transformation feeding a robotic-arm motion-planning routine for pick-and-place operations.

Achievements & Exposure

- **CODISSIA Intec / Electrotec:** exhibited the AMR prototype to industry visitors, demonstrating live navigation and monitoring.
- **Pitch Fest, Anna University:** pitched the AMR under the Emerging Startup category.
- **BITHACK Season 1:** First Prize.
- **Cadethon:** Most Innovative Design.

Education

B.E. Mechatronics Engineering

Expected 2028

Bannari Amman Institute of Technology, Tamil Nadu • CGPA: 8.67/10 • NPTEL Certification – Mechatronics